

ZOMATO DATASET ANALYSIS

InternSpark Data Science Internship

Student Details

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Internship Role: Data Science Intern

Project Title: Zomato Dataset Analysis

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

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Introduction

Food delivery and restaurant recommendation platforms generate large amounts of customer and restaurant data every day. Analyzing this data helps businesses understand customer preferences, restaurant performance, and market trends. In this project, the Zomato dataset was analyzed to gain insights into restaurant ratings, popular cuisines, location-wise restaurant distribution, pricing patterns, and online ordering trends.

The main objective of this analysis was to identify factors that influence restaurant ratings and customer engagement. Various data cleaning, preprocessing, and visualization techniques were applied using Python and Google Colab. The findings from this analysis can help businesses make better decisions regarding partnerships, marketing strategies, and customer experience improvements.

Dataset Overview

The dataset contains information about restaurants listed on the Zomato platform. It includes restaurant names, locations, cuisines, ratings, votes, online ordering availability, table booking options, and pricing details.

Dataset Details

Number of Records: 56,252

Number of Columns: 13

Key Attributes

- Address
- Restaurant Name
- Online Order
- Table Booking
- Rating
- Votes
- Phone Number
- Location
- Restaurant Type
- Dish Liked
- Cuisines
- Approximate Cost for Two People
- Listed Category

The dataset provides valuable insights into restaurant performance, customer preferences, and food industry trends.

Data Cleaning and Preparation

The dataset was cleaned and prepared using Python in Google Colab before performing the analysis.

The following preprocessing steps were completed:

- Imported and inspected the dataset.
- Checked for missing values.
- Filled missing values in categorical columns.
- Cleaned the Rating column and converted it into numeric format.
- Converted pricing information into numeric format.
- Removed inconsistent and invalid records.
- Standardized categorical data for analysis.
- Verified the dataset quality before visualization and modeling.

	0
address	17
name	16
online_order	19
book_table	58
rate	7838
votes	78
phone	1296
location	126
rest_type	338
dish_liked	28225
cuisines	203
approx_cost(for two people)	521
listed_in(type)	4610

dtype: int64

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 56252 entries, 0 to 56251
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   address                               56235 non-null  object
1   name                                  56236 non-null  object
2   online_order                          56233 non-null  object
3   book_table                            56194 non-null  object
4   rate                                  48414 non-null  object
5   votes                                 56174 non-null  object
6   phone                                 54956 non-null  object
7   location                              56126 non-null  object
8   rest_type                             55914 non-null  object
9   dish_liked                           28027 non-null  object
10  cuisines                              56049 non-null  object
11  approx_cost(for two people)           55731 non-null  object
12  listed_in(type)                       51642 non-null  object
dtypes: object(13)
memory usage: 5.6+ MB
```

Dataset Metrics

The following key metrics were calculated from the cleaned dataset.

Metric	Value
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Total Restaurants	8,792
Total Locations	94
Total Cuisine Types	2,724
Average Rating	3.14

These metrics provide an overview of the restaurant ecosystem represented in the dataset.

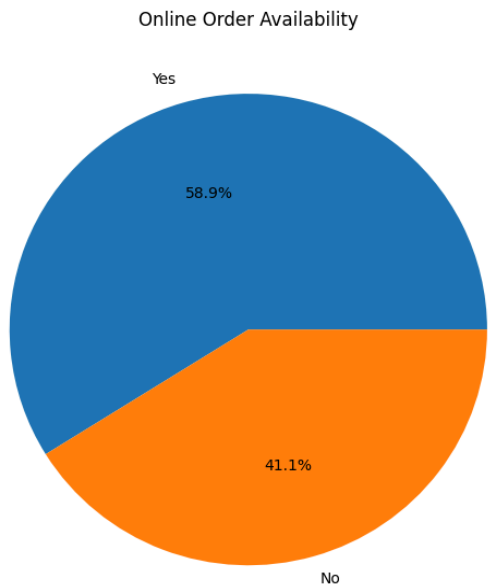
Total Restaurants: 11915
Total Locations: 2921
Total Cuisine Types: 5554
Average Rating: 3.17

Exploratory Data Analysis

Online Ordering Analysis

The availability of online ordering services was analyzed to understand digital adoption among restaurants.

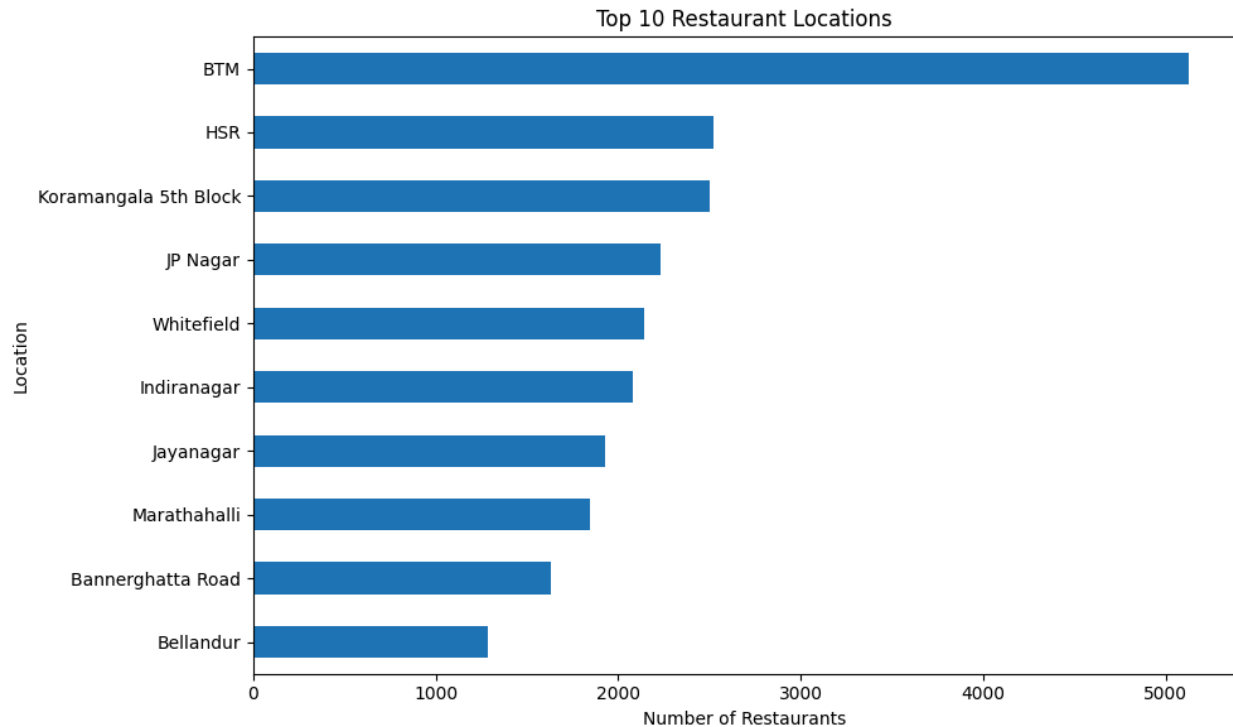
The analysis showed that a significant number of restaurants support online ordering, indicating the growing importance of digital food delivery services.



Location Analysis

Restaurant distribution was analyzed across different locations.

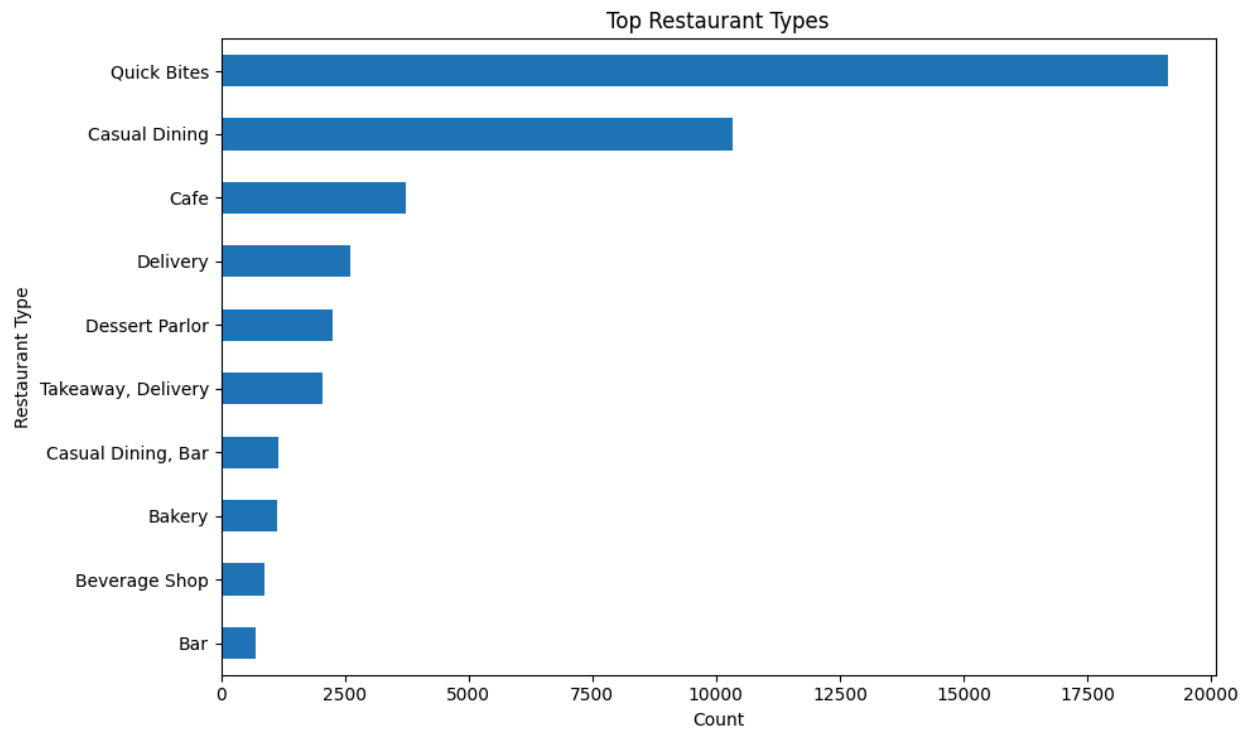
Certain locations contained a significantly higher number of restaurants compared to others, indicating strong business activity and customer demand in those areas.



Restaurant Type Analysis

Restaurant categories were analyzed to identify the most common types of dining establishments.

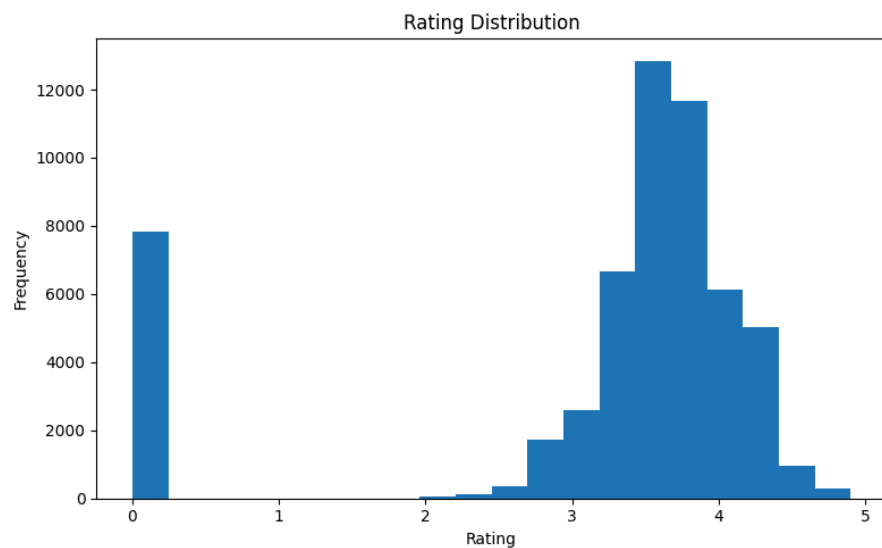
Quick-service restaurants and casual dining establishments represented a large portion of the dataset.



Rating Analysis

Customer ratings were analyzed to understand overall restaurant performance.

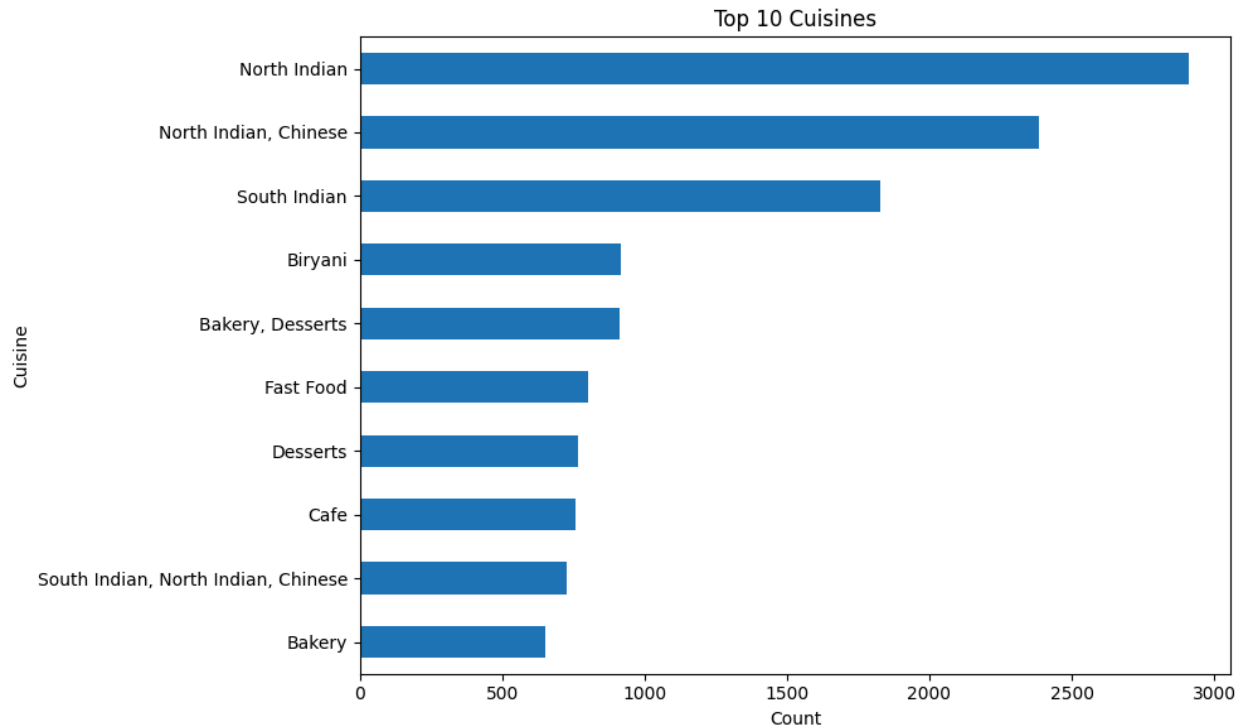
The rating distribution showed that most restaurants received ratings between 3 and 4.5, suggesting generally positive customer experiences.



Cuisine Analysis

Cuisine popularity was analyzed to identify customer preferences.

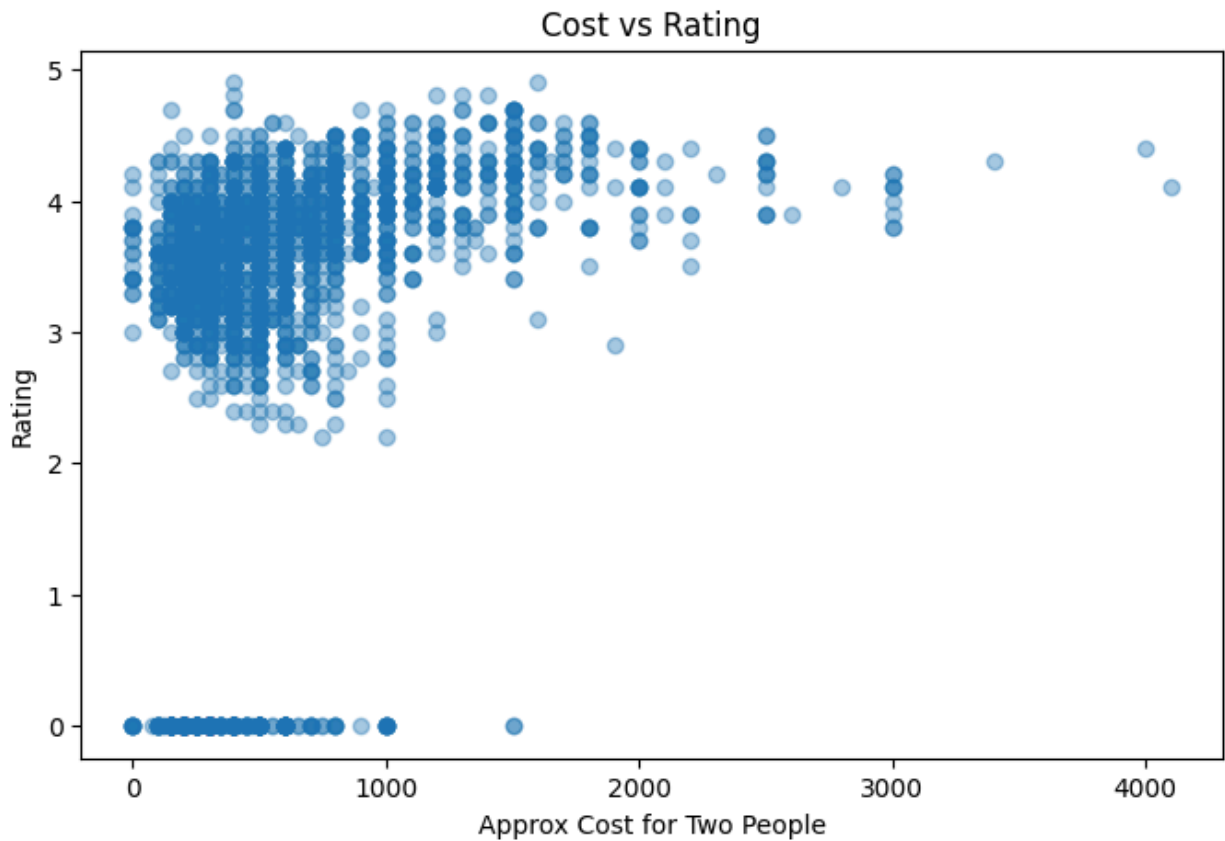
Several cuisine categories appeared frequently throughout the dataset, highlighting strong demand for those food types.



Cost vs Rating Analysis

The relationship between restaurant pricing and customer ratings was examined.

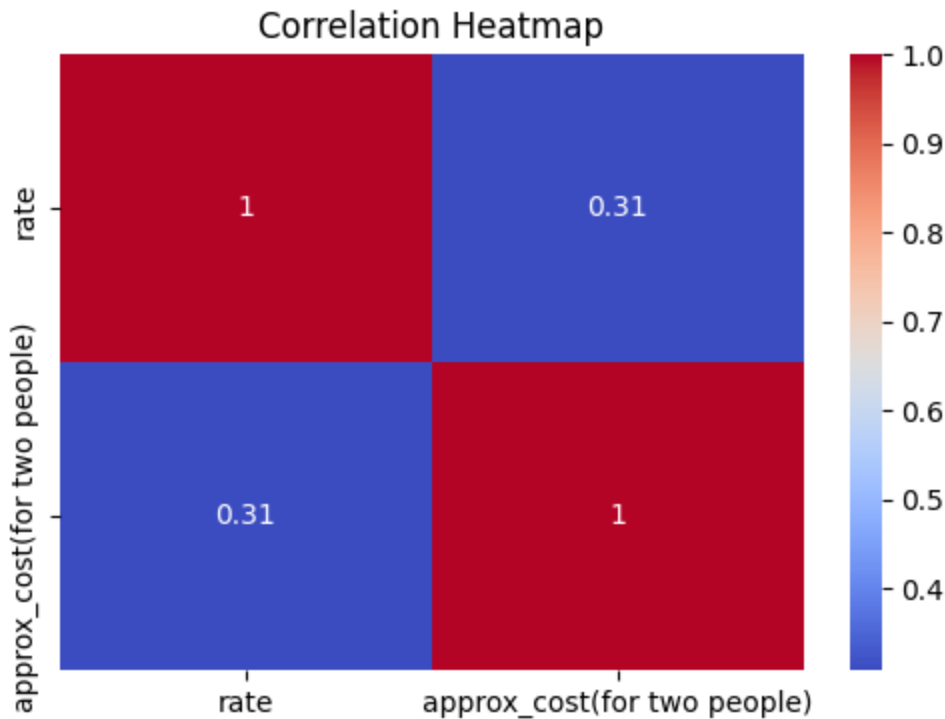
The analysis indicated that higher prices do not necessarily result in higher ratings. Customer satisfaction appears to depend on factors such as food quality, service quality, and overall dining experience.



Correlation Analysis

A heatmap was generated to study relationships among numerical variables.

The visualization helped identify the strength of relationships between ratings and restaurant pricing.



Key Findings

1. The dataset contains information on more than 56,000 restaurant records.
2. A total of 94 unique restaurant locations were identified.
3. Restaurants offered over 2,700 unique cuisine combinations.
4. The average restaurant rating was 3.14.
5. Online ordering is widely adopted across restaurants.
6. Restaurant businesses are concentrated in a few major locations.
7. Some cuisine categories dominate customer preferences.
8. Most restaurant ratings fall within the moderate-to-high range.
9. Pricing alone does not strongly influence restaurant ratings.
10. Digital ordering continues to play an important role in customer engagement.

Recommendations

Based on the analysis, the following recommendations are suggested:

1. Partner with highly rated restaurants and promote them prominently on the platform.

2. Focus marketing campaigns on popular cuisine categories to improve customer engagement.
3. Encourage restaurants to adopt online ordering and table booking facilities.
4. Use location-based marketing strategies in high-performing restaurant hubs.
5. Improve restaurant profile quality through better images, menus, customer reviews, and promotional content.

Executive Summary

This project focused on analyzing the Zomato restaurant dataset to understand customer preferences and restaurant performance. The dataset contained information about restaurant locations, cuisines, ratings, online ordering options, and pricing details. Before analysis, the data was cleaned by handling missing values and converting important fields into suitable formats.

Several visualizations were created to explore restaurant distribution, cuisine popularity, rating patterns, and online ordering trends. The analysis showed that restaurants are concentrated in specific locations and that online ordering is widely adopted across the platform. Popular cuisine categories appeared frequently throughout the dataset, indicating strong customer demand.

The relationship between pricing and ratings was also examined. The results suggested that higher prices do not necessarily guarantee better customer ratings. Instead, customer satisfaction appears to depend on multiple factors such as service quality, food quality, convenience, and overall dining experience.

Based on the findings, recommendations were provided to help improve restaurant visibility, customer engagement, and business growth. The project demonstrates how data analytics can be used to generate meaningful business insights and support decision-making in the food and restaurant industry.

Conclusion

The Zomato Dataset Analysis successfully provided valuable insights into restaurant trends, customer preferences, and rating patterns. Through data cleaning, exploratory analysis, and visualization, important relationships between locations, cuisines, pricing, and ratings were identified.

The analysis revealed that online ordering plays a significant role in customer engagement and that certain cuisines and locations attract a larger share of restaurants and customers. It was also observed that restaurant ratings are influenced by factors beyond pricing alone.

Overall, this project highlights the importance of data-driven decision-making in the restaurant industry. The insights obtained from the analysis can help businesses improve customer

satisfaction, strengthen restaurant partnerships, and develop more effective marketing strategies. Future analysis with additional customer review data could provide even deeper insights into consumer behavior and restaurant performance.